# Neil Yang

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## Education

**University of California, San Diego** — La Jolla, CA
*B.S. in Computer Science* — *Expected June 2028*

## Experience

**Esri** — Jun 2026 – Sep 2026
*Software Engineer Intern | ArcGIS Geodatabase Team* — *Redlands, CA*

- Designed a **splice-graph data model** for fiber-network associations, preserving topology across **500+ network edits**.
- Extended the **C++ network-index layer** for connectivity lookups, reducing fiber asset retrieval time by **10%** for operators.
- Optimized the **Python trace pipeline**, reducing utility-network query latency **25%** for faster splice diagnostics and planning.
- Compressed ArcGIS spatial metadata by **3 KB per asset**, lowering geodatabase storage and data transfer overhead.
- Shipped **TypeScript graph APIs** exposing fiber splice relationships for **1k+ downstream utility integrations**.

**Hornblower Group** — Jun 2025 – Sep 2025
*Software Engineer Intern | Payment SDK and Dashboard* — *Westlake Village, CA*

- Built an **AWS Lambda data-warehouse pipeline** converting **100K+ SQL-schema records** into DynamoDB JSON items.
- Developed a **reporting API** storing compact JSON in DynamoDB and signed PDFs in **Amazon S3** for **1k+ reports**.
- Deployed **Terraform-managed AWS APIs** for **300K+ users**, supporting a **$25M partner payment integration**.
- Debugged a **payment race condition** in Datadog, recovering **$4.5K+ in monthly revenue** from failed transactions.
- Built **GraphQL gift-card validation** for **50+ global clients**, blocking **100+ fraudulent transactions** to protect revenue.

**Google Mentorship Program (G-SWEP)** — Jan 2025 – Apr 2025
*Software Engineering Mentee* — *La Jolla, CA*

- Selected in the **top 10% of 400+ applicants**; built a **Python RAG pipeline on GCP** for grounded AI recommendations.

## Selected Software Projects

**MiniAWS – Distributed Object Storage** | *C++, Multithreading, TCP/IP, Linux*

- Engineered a multithreaded **C++ object store over TCP/IP**, supporting **4 concurrent remote storage clients**.
- Implemented **RAID 0 striping and asynchronous disk I/O**, increasing distributed object-store throughput **4×**.
- Reduced remote **write latency to <0.1 ms** with asynchronous I/O, accelerating persistence for storage clients at scale.
- Built a **bitmap metadata index** with **O(1) chunk lookup**, avoiding repeated scans for striped object retrieval on disk.
- Added **TCP acknowledgments and mutex-protected worker queues**, avoiding duplicate writes across **200+ operations**.

**RateMyDiningHall** | *TypeScript, GraphQL, Redis, MongoDB, Vercel*

- Launched a **TypeScript/GraphQL data platform** used by **300+ students** for searchable dining menus and reviews.
- Integrated **Redis caching with MongoDB**, maintaining **∼100 ms loads** across **1K+ menu items** for faster lookups.
- Designed **O(1) rating pre-aggregation**, eliminating redundant database calculations across menu and review queries.

**Manny.ai – AI CAD Copilot** | *Python, OpenAI, React, CadQuery*

- Built a **Python/OpenAI CAD copilot** with automated error correction, achieving **95% successful 3D edits** for designers.
- Compressed **3D mesh context into graphs**, reducing LLM input size **80%** to accelerate targeted editing and inference.

**VisualLink – Accessibility Computer Vision** | *Python, OpenCV, MediaPipe*

- Won **1st place and $500** at SacHacks VI for hands-free computer controls that eliminate specialized input costs for users.
- Reduced false-positive clicks by **85%** with OpenCV blink detection, improving real-time usability for mobility-impaired users.

## Technical Skills

**Languages:** C++, Python, Java, TypeScript, JavaScript, SQL, C
**Distributed/Data:** Multithreading, TCP/IP, Linux, Distributed Storage, DynamoDB, MongoDB, Redis, PostgreSQL
**Cloud/Tools:** AWS Lambda, Amazon S3, Terraform, REST APIs, GraphQL, Docker, CI/CD, Datadog, Git, GCP